

DDA3020 Machine Learning

Tutorial: Overfitting, Underfitting, and Bias–Variance Trade-off

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Instructions

This exercise sheet covers the main ideas of overfitting, underfitting, model complexity, and the bias–variance trade-off. You should be able to explain the difference between underfitting and overfitting, identify them from training and test errors, understand how model complexity affects training error and test error, and use the bias–variance decomposition to interpret the expected test error under squared loss.

Throughout this exercise sheet, input vectors are written in boldface, for example $\mathbf{x}$, $\mathbf{x}_i$, and $\mathbf{x}^{(n)}$. Labels or responses such as y remain scalar and are not bolded.

Useful Formulas and Concepts

1. Training dataset.

$$D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n.$$

2. Learned hypothesis.

Given a learning algorithm A , the hypothesis learned from dataset D is

$$h_D = A(D).$$

3. Regression data-generating model.

$$y = t(\mathbf{x}) + e, \quad e \sim N(0, \sigma^2),$$

where $t(\mathbf{x})$ is the unknown target function and e is the noise term.

4. Expected hypothesis function.

$$\bar{h}(\mathbf{x}) = \mathbb{E}_D [h_D(\mathbf{x})].$$

5. Expected test error under squared loss.

$$\mathbb{E}_{(\mathbf{x}, y), D} \left[(h_D(\mathbf{x}) - y)^2 \right].$$

6. Bias–variance decomposition.

$$\begin{aligned} \mathbb{E}_{(\mathbf{x}, y), D} \left[(h_D(\mathbf{x}) - y)^2 \right] &= \mathbb{E}_{(\mathbf{x}, y), D} \left[(h_D(\mathbf{x}) - \bar{h}(\mathbf{x}))^2 \right] \\ &\quad + \mathbb{E}_{(\mathbf{x}, y)} \left[(\bar{h}(\mathbf{x}) - t(\mathbf{x}))^2 \right] \\ &\quad + \mathbb{E}_{(\mathbf{x}, y)} \left[(t(\mathbf{x}) - y)^2 \right]. \end{aligned}$$

7. Variance term.

$$\mathbb{E}_{(\mathbf{x},y),D} \left[\left(h_D(\mathbf{x}) - \bar{h}(\mathbf{x}) \right)^2 \right].$$

This term measures how much the learned hypothesis changes when trained on different datasets.

8. Bias squared term.

$$\mathbb{E}_{(\mathbf{x},y)} \left[\left(\bar{h}(\mathbf{x}) - t(\mathbf{x}) \right)^2 \right].$$

This term measures how far the average prediction is from the true target function.

9. Noise term.

$$\mathbb{E}_{(\mathbf{x},y)} \left[\left(t(\mathbf{x}) - y \right)^2 \right].$$

This term is caused by the intrinsic randomness in the data-generating process and cannot be eliminated by changing the learning algorithm.

- 10. Model complexity.** As model complexity increases, training error usually decreases. However, test error may first decrease and then increase. Low-complexity models tend to have high bias and low variance; high-complexity models tend to have low bias and high variance.

Notation Reminder

We write $\mathbf{x} \in \mathbb{R}^d$ for an input vector and $y \in \mathbb{R}$ for a scalar response. A training dataset is

$$D = \{(\mathbf{x}_i, y_i)\}_{i=1}^n.$$

The learned model from D is denoted by h_D . The unknown target function is denoted by $t(\mathbf{x})$. The expected hypothesis function over possible training datasets is

$$\bar{h}(\mathbf{x}) = \mathbb{E}_D [h_D(\mathbf{x})].$$

For a one-dimensional example, a scalar input may be written as x , but all general input vectors are written in boldface.

Exercises

Q1. Basic Concepts

Answer the following questions briefly.

- (a) What does underfitting mean in terms of training error?
- (b) What does overfitting mean in terms of training error and test error?
- (c) Give one possible reason for underfitting.
- (d) Give one possible reason for overfitting.
- (e) Give one remedy for underfitting and one remedy for overfitting.

Q2. Model Complexity in Polynomial Regression

Suppose the true relationship between $\mathbf{x}$ and y is nonlinear. We fit the same training dataset using the following models:

- Model 1: a linear regression model;
- Model 2: a 2nd-order polynomial regression model;
- Model 3: a 9th-order polynomial regression model.

Answer the following questions.

- (a) Which model is most likely to underfit?
- (b) Which model is most likely to overfit?
- (c) Which model is most likely to have a better balance between underfitting and overfitting?
- (d) Explain your answers using model complexity.

Q3. Training Error and Test Error

A model is trained and tested on a regression task. The following table gives the training error and test error.

Model	Training Error	Test Error
A	8.2	8.7
B	1.1	7.9
C	3.4	3.8
D	6.9	7.2

Answer the following questions.

- (a) Which model is most likely to be overfitting?
- (b) Which models are more likely to be underfitting?
- (c) Which model would you choose based on test error?
- (d) Explain why training error alone is not sufficient for model selection.

Q4. Polynomial Order and Test Error

A polynomial regression model is trained with different polynomial orders. The training and test errors are shown below.

Polynomial Order	Training Error	Test Error
1	10.4	11.1
2	5.6	6.1
4	3.2	3.6
6	1.8	5.4
9	0.4	9.8

Answer the following questions.

- (a) Describe the trend of training error as the polynomial order increases.
- (b) Which polynomial order gives the lowest test error?
- (c) Which polynomial order is most likely to underfit?
- (d) Which polynomial order is most likely to overfit?
- (e) Explain how this table illustrates the bias–variance trade-off.

Q5. Bias and Variance

For each statement, write **True** or **False**. Give a short explanation.

- (a) A high-bias model is usually too simple.
- (b) A high-variance model is usually very sensitive to the particular training dataset.
- (c) Increasing model complexity usually increases bias.
- (d) Increasing model complexity usually increases variance.
- (e) Noise can be completely removed by choosing a sufficiently complex model.

Q6. Bias–Variance Decomposition

In the lecture, the data are modeled by

$$y = t(\mathbf{x}) + e,$$

where

$$e \sim N(0, \sigma^2).$$

For a learning algorithm A , the learned hypothesis from dataset D is

$$h_D = A(D),$$

and the expected hypothesis is

$$\bar{h}(\mathbf{x}) = \mathbb{E}_D [h_D(\mathbf{x})].$$

The expected test error can be decomposed as

$$\begin{aligned} \mathbb{E}_{(\mathbf{x}, y), D} \left[(h_D(\mathbf{x}) - y)^2 \right] &= \mathbb{E}_{(\mathbf{x}, y), D} \left[(h_D(\mathbf{x}) - \bar{h}(\mathbf{x}))^2 \right] \\ &\quad + \mathbb{E}_{(\mathbf{x}, y)} \left[(\bar{h}(\mathbf{x}) - t(\mathbf{x}))^2 \right] \\ &\quad + \mathbb{E}_{(\mathbf{x}, y)} \left[(t(\mathbf{x}) - y)^2 \right]. \end{aligned}$$

Identify the following terms:

- (a) the variance term;
- (b) the bias squared term;
- (c) the noise term;
- (d) why the noise term cannot be eliminated by changing the learning algorithm.

Q7. Numerical Bias–Variance Calculation

Suppose we repeatedly sample training datasets D from the same distribution and train a regression model h_D . For a one-dimensional scalar test sample

$$(x, y) = (5, 10),$$

suppose the true target value is

$$t(5) = 9.5.$$

The predictions of 10 trained models on $x = 5$ are

$$9, 11, 23, 6, 8, 12, 10, 4, 13, 7.$$

Using these 10 predictions, compute:

- (a) the empirical average prediction $\bar{h}(5)$;
- (b) the empirical bias squared $(\bar{h}(5) - t(5))^2$;
- (c) the empirical variance

$$\frac{1}{10} \sum_{i=1}^{10} (h_{D_i}(5) - \bar{h}(5))^2;$$

- (d) the empirical MSE

$$\frac{1}{10} \sum_{i=1}^{10} (h_{D_i}(5) - y)^2.$$

Q8. Learning Curves

For the same model, suppose the number of training instances increases. Consider the following two regimes.

- **Regime 1:** The training error is much lower than the test error. The training error is below an acceptable error level ϵ , but the test error is above ϵ .
- **Regime 2:** The training error is above the acceptable error level ϵ .

Answer the following questions.

- (a) Which regime indicates high variance?
- (b) Which regime indicates high bias?
- (c) Give one remedy for Regime 1.
- (d) Give one remedy for Regime 2.

Q9. Decision Trees and Random Forests

According to the lecture, single decision trees and random forests have different bias–variance behaviors.

- (a) A single pruned tree usually has high bias or high variance?
- (b) A single deep tree usually has high bias or high variance?
- (c) Why can random forests reduce variance compared with a single tree?
- (d) Does a random forest necessarily reduce bias? Answer yes or no, and explain briefly.

Q10. Choosing a Suitable Model

Suppose you are given the following three candidate models for a regression task.

Model	Model Complexity	Training Error	Test Error
A	Low	12.0	12.8
B	Medium	4.5	5.0
C	High	0.8	9.6

Answer the following questions.

- (a) Which model is most likely underfitting?
- (b) Which model is most likely overfitting?
- (c) Which model would you choose?
- (d) Explain the choice using bias and variance.

Solutions

Q1. Answer

- (a) Underfitting means that the model cannot predict the labels of the training data well. In terms of error, the training error is high.
- (b) Overfitting means that the model predicts very well on the training data but poorly on the test data. In terms of error, the training error is low, but the test error is much higher.
- (c) One possible reason for underfitting is that the model is too simple for the data. For example, a linear model may underfit nonlinear data.
- (d) One possible reason for overfitting is that the model is too complex for the amount of training data. For example, a very high-order polynomial model may fit noise in the training data.
- (e) A remedy for underfitting is to use a more complex model or add more informative features. A remedy for overfitting is to reduce model complexity or add more training instances.

Q2. Answer

- (a) Model 1 is most likely to underfit because a linear model is too simple to capture a nonlinear relationship.
- (b) Model 3 is most likely to overfit because a 9th-order polynomial model is highly flexible and can fit accidental patterns or noise in the training data.
- (c) Model 2 is most likely to have a better balance because a 2nd-order polynomial model is more flexible than a linear model but less extreme than a 9th-order polynomial model.
- (d) Model complexity increases from Model 1 to Model 3. A model with too little complexity may have high bias and underfit. A model with too much complexity may have high variance and overfit. A moderate-complexity model is often preferred.

Q3. Answer

- (a) Model B is most likely to be overfitting. It has very low training error, 1.1, but much higher test error, 7.9. The large gap indicates overfitting.
- (b) Models A and D are more likely to be underfitting. Both have high training errors and high test errors, which suggests that the models are too simple or the features are not informative enough.
- (c) Model C should be chosen based on test error because it has the lowest test error:

3.8.

- (d) Training error alone is not sufficient because a model can achieve very low training error by overfitting the training data. The goal is prediction on unseen test data, so test error is more relevant for model selection.

Q4. Answer

- (a) The training error decreases as the polynomial order increases:

$$10.4 \rightarrow 5.6 \rightarrow 3.2 \rightarrow 1.8 \rightarrow 0.4.$$

This is expected because higher-order polynomial models are more flexible.

- (b) Polynomial order 4 gives the lowest test error:

$$3.6.$$

- (c) Polynomial order 1 is most likely to underfit because both its training error and test error are high.
- (d) Polynomial order 9 is most likely to overfit because it has very low training error but very high test error.
- (e) As model complexity increases, bias tends to decrease, so training error decreases. However, variance tends to increase, so test error may increase after a certain point. The test error is lowest at an intermediate complexity, which illustrates the bias-variance trade-off.

Q5. Answer

- (a) **True.** A high-bias model is usually too simple and cannot capture the true structure of the data.
- (b) **True.** A high-variance model changes significantly when trained on different training datasets.
- (c) **False.** Increasing model complexity usually decreases bias because the model becomes more flexible.
- (d) **True.** Increasing model complexity usually increases variance because the model becomes more sensitive to the training dataset.
- (e) **False.** Noise is intrinsic to the data distribution and feature representation. It cannot be completely removed by choosing a more complex model.

Q6. Answer

- (a) The variance term is

$$\mathbb{E}_{(\mathbf{x}, y), D} \left[\left(h_D(\mathbf{x}) - \bar{h}(\mathbf{x}) \right)^2 \right].$$

It measures how much the learned hypothesis changes across different training datasets.

- (b) The bias squared term is

$$\mathbb{E}_{(\mathbf{x}, y)} \left[\left(\bar{h}(\mathbf{x}) - t(\mathbf{x}) \right)^2 \right].$$

It measures how far the average hypothesis is from the target function.

- (c) The noise term is

$$\mathbb{E}_{(\mathbf{x}, y)} \left[\left(t(\mathbf{x}) - y \right)^2 \right].$$

Since $y = t(\mathbf{x}) + e$, this term comes from the random noise e .

- (d) The noise term cannot be eliminated by changing the learning algorithm because it is caused by the intrinsic randomness in the data-generating process. Even the true target function $t(\mathbf{x})$ cannot perfectly predict y when $e \neq 0$.

Q7. Answer

The 10 predictions are

$$9, 11, 23, 6, 8, 12, 10, 4, 13, 7.$$

- (a) The empirical average prediction is

$$\bar{h}(5) = \frac{9 + 11 + 23 + 6 + 8 + 12 + 10 + 4 + 13 + 7}{10} = \frac{103}{10} = 10.3.$$

- (b) Since $t(5) = 9.5$, the empirical bias squared is

$$(\bar{h}(5) - t(5))^2 = (10.3 - 9.5)^2 = 0.8^2 = 0.64.$$

- (c) The empirical variance is

$$\frac{1}{10} \sum_{i=1}^{10} (h_{D_i}(5) - \bar{h}(5))^2.$$

Substituting $\bar{h}(5) = 10.3$, we get

$$\begin{aligned} \text{Variance} &= \frac{1}{10} [(9 - 10.3)^2 + (11 - 10.3)^2 + (23 - 10.3)^2 \\ &\quad + (6 - 10.3)^2 + (8 - 10.3)^2 + (12 - 10.3)^2 + (10 - 10.3)^2 \\ &\quad + (4 - 10.3)^2 + (13 - 10.3)^2 + (7 - 10.3)^2] \\ &= 24.81. \end{aligned}$$

- (d) Since $y = 10$, the empirical MSE is

$$\frac{1}{10} \sum_{i=1}^{10} (h_{D_i}(5) - 10)^2.$$

Therefore,

$$\begin{aligned} \text{MSE} &= \frac{1}{10} [(9 - 10)^2 + (11 - 10)^2 + (23 - 10)^2 + (6 - 10)^2 \\ &\quad + (8 - 10)^2 + (12 - 10)^2 + (10 - 10)^2 + (4 - 10)^2 + (13 - 10)^2 + (7 - 10)^2] \\ &= \frac{1}{10} (1 + 1 + 169 + 16 + 4 + 4 + 0 + 36 + 9 + 9) \\ &= \frac{249}{10} = 24.9. \end{aligned}$$

Q8. Answer

- (a) Regime 1 indicates high variance. The training error is much lower than the test error, which is a symptom of overfitting.
- (b) Regime 2 indicates high bias. The training error is above the acceptable error level ϵ , which means the model is not powerful enough to fit the training data well.
- (c) One remedy for Regime 1 is to add more training instances. Another remedy is to reduce model complexity.
- (d) One remedy for Regime 2 is to add more features. Another remedy is to use a more complex model, such as a nonlinear or kernel model.

Q9. Answer

- (a) A single pruned tree usually has high bias and low variance. It may underfit because pruning restricts model complexity.
- (b) A single deep tree usually has low bias and high variance. It may overfit because it can fit the training data very closely.
- (c) Random forests introduce data and model randomness by training many different trees and averaging their predictions. Averaging reduces the sensitivity to any particular training dataset, so the variance is reduced compared with a single tree.
- (d) No. A random forest does not necessarily reduce bias. The trees are trained independently and the overall model complexity is not necessarily increased in a way that guarantees lower bias.

Q10. Answer

- (a) Model A is most likely underfitting. It has low complexity and both training error and test error are high.
- (b) Model C is most likely overfitting. It has very low training error but high test error.
- (c) Model B should be chosen because it has the lowest test error:

5.0.

- (d) Model A likely has high bias because it is too simple. Model C likely has high variance because it is too complex and fits the training data too closely. Model B has a better balance between bias and variance, giving the best test performance among the three models.